

MediSophos: Multilingual QR System for Medicine Info

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Abstract—In our daily lives, we often rely on medications for various reasons. These medicines come with essential information like medicine manufacture and expiry date, which is very small and sometimes partially erased. This situation often leads to the consumption of expired medicine, which can cause various side effects. In order to prevent this, we propose a solution that integrates QR codes with key information such as manufacture and expiry dates with usage info. Additionally, it contains a multilanguage switcher, which helps people view it in their preferred regional language, ensuring accessibility for everyone. This user-friendly interface helps people to simply scan a QR code and instantly access medicine details, helping them consume proper medications.

Index Terms—QR code, expiry date detection, multi-language support, Patient safety, usage info, Medicine management.

I. INTRODUCTION

In today's healthcare sector, the Medicine management is an important process in hospitals, pharmacies, and healthcare centres to ensure that patients receives genuine and safe medicines. Traditional records keeping methods are often manual, time consuming, leads some error, and creates problem such as expired medicine distribution and lack of tracking of stock levels. So the need for simple, affordable, and reliable digital solution is important.

The advancement of QR code scanner technology to manage the entire medicine lifecycle from entry to dispensing. QR code is fast and reliable method to store and collect essential information throughout scanning. They have proven effective in other various domains like retail, logistics, goods services but not in medicines that can be unique to identify, tracked, and verify Ref. [1]

The Existing systems for pharmacies tracking mainly depends on cost solution such RFID tags or blockchain- based supply chains. while these technologies enhance security and traceability, they are often expensive, require technical experts , and are not suitable for small pharmacies or rural healthcare centres. Other simpler solutions, like barcode-based systems, are limited by storage capacity and scanning reliability. Therefore, there is a clear gap for lightweight, cost-

effective, and scalable system that combines the accessibility of QR technology with efficient data management Ref. [2] [3] The proposed MediSophos- Medicine Management System using QR code addresses these gaps by automatic medicine tracking, stock monitoring, and expiry management in a single platform. Each medicine is give a unique QR code generated from its details, which can be scanned to instantly retrieve information such as name, manufacture and expiry dates, and manufacturers details. This system not only simplifies inventory handling for pharmacists but also leads patients to view and verify the safety of their medicines. And this included offline-based approach that allow scanning essential medicine data even without internet connectivity. This ensures uninterrupted service in rural healthcare centres and remote areas where consistent network access is a challenge. The system securely stores and syncs data once connectivity is restored, maintaining integrating and reliable. In addition a language-switching feature that allows users to choose their preferred regional language. This makes the platform easier to use for people from different linguistic backgrounds, including patients, pharmacists, and healthcare workers. by supporting multi language, the system ensures that everyone can access and understand medicine information without confusion. This system has the potential to improve transparency, reduce human error, secure, user-friendly, and ensure trust in pharmaceutical management across healthcare.

II. LITERATURE REVIEW

Explored several research studies on the use of QR codes in healthcare to improve medicine tracking, patient safety, and data accessibility. [4] R. Liu et al., EQRC - Supporting Online and Offline Transactions. This paper introduces a secure QR code framework. This QR code method is designed to encrypt and can be verified without constant internet availability.

This concept shows how offline authentication can be safely managed. [1] J. Thobias and A. Kiwanuka, Implementation of an M-Health Data model for Improving health information access. This paper shows the uses of SMS-based technology for the improvement of healthcare access in low-resource places. This method sends health information to users through text messages, allowing people with the unavailability of smartphones or internet access to receive important medical

data. [5] R. M. Nor et al., ScanMed-A mobile medicine adherence application with intake validation using QR code. In this study a mobile application that helps patients follow their medication schedule. uses QR codes printed on medicine packages to confirm a patient has taken their medicine according to the dose. Ref. [6] A. K. Saw et al.,- A novel approach for seamless reminder apps for expiry date with QR. This paper focuses on developing a reminder application that helps users track the expiry dates of products, including medicines, by scanning and provides alerts through an intelligent reminder interface and details. [7] P. K. Pal and Kule- A new approach to pharmaceutical product verification using barcode and QR code. It is a verification framework that uses both barcodes and QR codes to identify and verify pharmaceutical products. This makes it easier for users to check medicine details. [2] S. U. Mathaba-Interfacing RFID and Web 2.0 technologies to improve inventory management. It shows how integrating digital tools can help monitor supplies efficiently. [8] M. M. Aulia et al- Protected QR code-based anti-counterfeit system for pharmaceutical Manufacturing. This work proposes a secure QR code system designed to prevent counterfeit medicines in the pharmaceutical industry. It uses encrypted QR codes that store unique product details to ensure each medicine. [3] H. A. M. Wahsheh-Secure and usable QR codes focus on balancing security and usability in QR code design.

III. METHODOLOGY

The methodology adapted for this work focuses on developing a user friendly and reliable system which is capable of displaying essential medicine information through QR code scanning. This process involves several stages, such as system design, data entry, QR code generation, scanning and retrieval, language switcher and also future offline expand. The stages included in this process ensures the accuracy, accessibility of the proposed system.

A. System Architecture Design

The proposed system architecture follows a model which stores each medicine's critical details, such as its name, product number, manufacturing date, expiry date and other information. Each record is assigned with a unique QR code through a standardized algorithm compliant with ISO/IEC 18004 specifications. Which is very similar approach that have been implemented in Scanned, where QR codes were used for intake validation in medical application. [4]

B. Data Entry and QR code generation

The users input medicine data through a secure interface. Once fetched, the data is stored in a centralized database, and a QR code is generated automatically for entry. This system applies similar techniques to those proposed by Aulia et al Ref. [5], which also utilizes the secure QR generation.

C. Scanning and Data Retrieval

When the user scans a QR code via mobile camera, the encoded data is decoded and mapped to the corresponding record in the database. Unlike the traditional models that rely on redirecting users to websites, this system displays the information directly through Google's native scanning. This retrieval process aligns with methods described by pal and kule [6]

D. Multilingual Support

To promote inclusivity, a dynamic language-switching module is also integrated into the system. Through API-based translation such as Google Cloud Translation, users can instantly switch the displayed information to their preferred language. This feature aligns with principles in Thobias and Kiwanuka's study on m-health data models, while extending the idea of embedding live translation functionality directly into QR scanning workflow. [8]

E. Offline SMS-Based Access(Future Scope)

In areas with limited or no internet connectivity, this system proposes a enhancement through an SMS -based retrieval mechanism. This concept draws inspiration from prior SMS based health systems such as those proposed by Thobias and Kiwanuka Ref. [8] and Jaiswal Ref. [7], which is focusing on healthcare notifications and general information. However, the novelty of this work lies in integrating the SMS-based retrieval directly with the same database used for online QR access, which enables real time synchronization between offline and online modes. Unlike earlier studies that used prestored message templates or manual data flow information from the database before sending it to the user. This ensures consistency of information between online and offline environments, offering a more scalable solution.

F. Testing and Validation

This system undergoes structured testing phases to ensure reliability and accuracy. Unit testing and integration testing are performed to verify the correct functioning of each module. These steps align with quality assessment practices used in similar healthcare informatics systems Ref. [1], while ensuring usability under both online and offline scenarios.

IV. EXISTING SYSTEM

In the existing systems used within the pharmaceutical industry, most QR code implementations are limited to either redirecting users to external web pages or mobile applications for retrieving product details. These systems depend heavily on stable internet connectivity and third-party platforms, making them inaccessible in low-network regions. Moreover, users are often required to install dedicated applications, which can be inconvenient and data-intensive, especially for rural populations.

Previous research, such as ScanMed Ref. [4], focused on QR-based medicine adherence tracking, while Aulia et al Ref. [5] and Pal and Kule Ref. [6] explored secure and authenticated pharmaceutical product verification using QR and barcode technology. Similarly, Thobias and Kiwanuka Ref. [8] and Jaiswal Ref. [7] proposed SMS-based healthcare information systems that primarily distributed static, predefined messages.

Limitations of existing approaches:

- 1 It is dependent on continuous internet access for QR data retrieval.
- 2 It Requires external websites or applications to display results.
- 3 Lack of multilingual support for users.
- 4 Offline systems operating on pre-stored or template-based data rather than real-time.

V. PROPOSED SYSTEM

A. Overview

This system focuses on a user-friendly way to verify medicine using QR code-based methods. Interface. Each medicine is labelled with a unique QR code generated by the manufacturer. And when the user scans, it displays information about the medicine, such as expiry date, usage, and other information. On scanning the QR code, the medicine's details are displayed directly in the Google interface without the need to open any external links or mobile applications, eliminating the need for third-party apps or web links. This ensures quick access to essential data. Additionally, this system has a unique feature that integrates multilingual support and user convenience, allowing users to view information in their preferred language. In future enhancements, this system aims to include an offline SMS based approach to retrieve medicine details even during the unavailability of the internet, which ensures the continuous functionality and easy accessibility in remote areas

B. System Architecture

The system architecture is designed around three major components: the user interface, the backend database, and the communication layer.

- The user interface is responsible for capturing the QR code, and the decoded data is processed and sent to the backend for verification. This can be possible not only by using Google Lens but also with the use of various other Visual Search Systems.
- The backend database stores verified medicine records, including the manufacturer name, manufacture date, expiry date, price and information on what the medicine does. It also ensures data integrity, authenticity and real-time retrieval.
- The communication layer connects the two modules through a secure API, which handles the user request, and the system responds efficiently. This ensures real-time verification against pharmaceutical regulatory databases.

C. Module Description

1. QR code scanning module: This module captures the QR code using the device's camera and decodes it using a QR recognition algorithm. Supports both static and dynamic codes, and it also ensures that even low-quality images can be interpreted correctly.
2. Database management module: With software like SQL Lite, it stores the key information about medicines, including product specifications and expiry dates. This helps in maintaining logs of user queries to support health monitoring systems.
3. Language Translation module: The module provides multi-language support, which allows users to view information in their local language. Relies on predefined translation mappings in future versions, and it also integrates with cloud translation APIs for updates.
4. Offline SMS Service module (Future scope): Specifically in areas where mobile internet connectivity is poor or unavailable, this module allows users to send the QR code data through SMS. This server will then reply with basic medicine details, which again ensures accessibility for all types of users.

D. Workflow

This diagram shows how this process works when a QR code is scanned, the encoded data is decoded and sent to the backend via a secure API where Data gets extracted and

verified and information gets displayed for future extension there is a SMS query module.

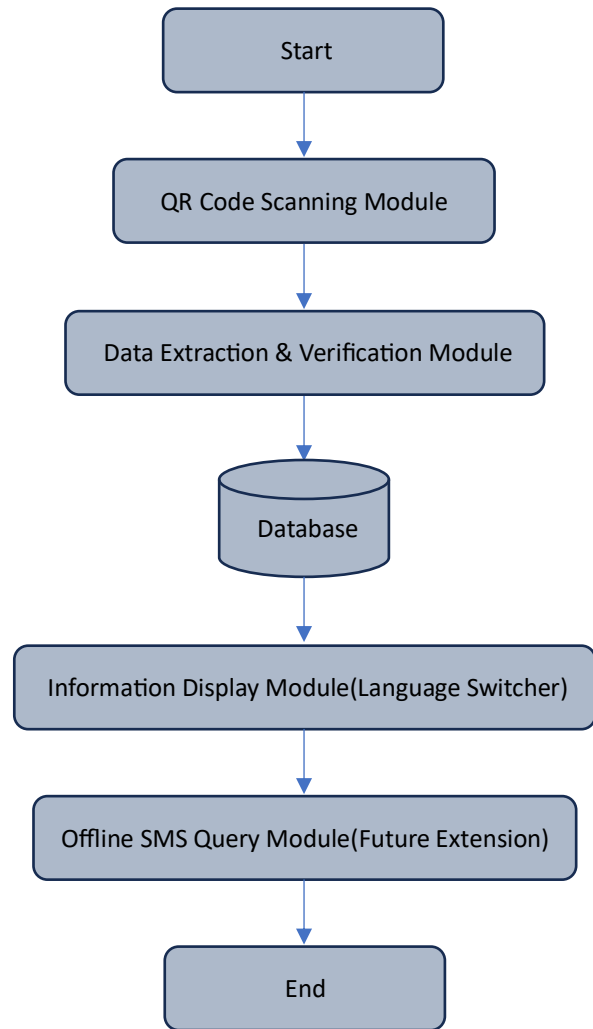


Fig. 1. Flowchart of the proposed QR Code-based Medicine Information System

E. Implementation Details

The prototype is implemented using A semi-web based QR code medicine management system built with use of HTML, CSS (or a framework like Flask, React) while the backend using database like MySQL, SQLite to store details. The QR code module generates and links each QR code to its stored data.

VI. FINDINGS

Following implementation and testing of the proposed system, the users could scan the QR code printed on the medicine package and immediately access the information about the medicine, such as the manufacture date, the expiry date and its application. Another advantage was that this system took the form of being able to view such information in several regional languages based on the choice of the user and this made the system more comprehensible to a larger audience. The interface is easy to navigate and is smooth and reliable to most of the people. Such a system assists in minimizing the possibility of taking expired medicine, and it is also easier to determine the expiry date even when the writing on the medicine strips are worn off. They did not need to squint to see on the package small or fading text. The feature can be utilized by old users and individuals with poor

vision. It also offers an easy to use solution to the common user of medicines.

Fig. 1. Flowchart of the proposed QR Code-based Medicine Information System.

VII. CONCLUSION

The proposed “MediSophos- Medicine Management System using QR code” offers an efficient and secure way to manage medical inventory and ensure drug authenticity. By integrating AI-based analysis, Cybersecurity encryption and an offline QR scanning module, it enhances reliability even in low connectivity areas as future scope. The additional feature multi language-switching interface improves accessibility for many users. This system minimises errors, prevents counterfeit distribution, and support smart decision making in healthcare. In advance integrating with blockchain and IOT can further strengthen transparency, making the solution adaptable for large scale medical systems.

VIII. FUTURE SCOPE

The MediSophos- Medicine Management System using QR code presents a strong foundation for developing a more smart and easy healthcare infrastructure. In future, the system can be enhanced to provide an easy connection between medicine manufactures, hospitals, pharmacies and patients through a same platform that gives easy in low network environments. The offline QR scanner can be extended into a hybrid mobile or web application that supports local data storage and automatic synchronisation when connectivity is restored, ensuring uninterrupted service in rural or remote areas. Another improvement is to focus on AI-module can be used to study usage patterns, predict medicine shortages, and detect unusual scanning activities that may indicate counterfeit medicines. This will help to maintaining proper stock levels and improves safety of patients. On security side, the system can include protection like secure login, realtime alerts for any unauthorised access for manufacturers, pharmaceutical. A major enhancement in this system that will be implemented is multilingual or language-switching interface, enabling users to interact with the system in their preferred regional language. This make the application more interactive and accessible to manufacturers , pharmacists and patients in rural and urban areas. This system help for users like pharmacists, hospital staff, manufacturers and for large scale use in smart healthcare and medicine supply management.

REFERENCES

- [1] J. Thobias and A. Kiwanuka, “Design and implementation of an m-health data model for improving health information access for reproductive and child health services in low resource settings using a participatory action research approach,” *BMC medical informatics and decision making*, vol. 18, no. 1, p. 45, 2018.
- [2] S. U. Mathaba, “Interfacing rfid and web 2.0 technologies to improve inventory management in south african enterprises,” Ph.D. dissertation, University of Zululand, 2014.
- [3] H. A. M. Wahsheh, “Secure and usable qr codes,” 2019.
- [4] R. Liu, J. Song, Z. Huang, and J. Pan, “Eqrc: A secure qr code-based e-coupon framework supporting online and offline transactions,” *Journal of Computer Security*, vol. 28, no. 5, pp. 577–605, 2020.
- [5] R. M. Nor, N. A. Mohamadali, K. Azmi, A. Marzuki, L. M. Nor, and M. Yusof, “Scanmed: A mobile medicine adherence application with intake validation using qr code,” in *2016 6th International Conference on Information and Communication Technology for The Muslim World (ICT4M)*. IEEE, 2016, pp. 112–117.
- [6] A. K. Saw, S. Kaushik, A. Sharma, and B. K. Nayak, “A novel approach for seamless reminder apps for expiry date with qr,” in *2024 3rd, International Conference on Artificial Intelligence For Internet of Things (AIIoT)*. IEEE, 2024, pp. 1–6.
- [7] P. K. Pal and M. Kule, “A new approach to pharmaceutical product verification using barcode and qr code,” in *International Conference on Frontiers in Computing and Systems*. Springer, 2022, pp. 223–229.
- [8] M. Masruk Aulia, N. Saha, and M. Mostafizur Rahman, “Protected qr code-based anti-counterfeit system for pharmaceutical manufacturing,” *arXiv e-prints*, pp. arXiv–2404.

- [1] J. Thobias and A. Kiwanuka, “Design and implementation of an m-health data model for improving health information access for reproductive and child health services in low resource settings using a participatory action research approach,” *BMC medical informatics and decision making*, vol. 18, no. 1, p. 45, 2018.
- [2] S. U. Mathaba, “Interfacing rfid and web 2.0 technologies to improve inventory management in south african enterprises,” Ph.D. dissertation, University of Zululand, 2014.
- [3] H. A. M. Wahsheh, “Secure and usable qr codes,” 2019.
- [4] R. Liu, J. Song, Z. Huang, and J. Pan, “Eqrc: A secure qr code-based e-coupon framework